WEEK 6 ASSIGNMENT: GIT, GITHUB & FINAL DATA ANALYTICS PROJECT

Final Capstone Project Practical Submission | 100 Marks Evaluation

Student Name	Ayush	**Assignment Module**	Week 6: Git, GitHub & Capstone Project
Degree & Branch	B.Tech in CSE (IoT)	**Evaluation Marks**	100 Marks (Tasks 1 to 8 Covered)
Institution	Prestige Institute of Engineering, Management & Research, Indore	**Core Technologies**	Python, Git, GitHub, Pandas, Seaborn, Power BI
GitHub Repository	https://github.com/Ayush-Raghuwanshi-Dev/DA_Internship.git	**Submission Date**	October 2026

Task 1: Git & GitHub Basics (10 Marks)

Question: *Set up Git and GitHub and perform the following: Install Git. Create a GitHub account if you do not already have one. Create a new repository. Clone the repository to your local system. Add project files to the repository. Create at least one commit. Push the changes to GitHub.*

Student Explanation & Implementation Summary:

- Verified Git binary on Windows (version 2.49.0.windows.1) configured with user identity.
- Synced with GitHub remote repository: https://github.com/Ayush-Raghuwanshi-Dev/DA_Internship.git on branch main.
- Initialized project tracking and verified staging index and working tree connectivity.

Output Screenshot (Git & GitHub Configuration Audit):

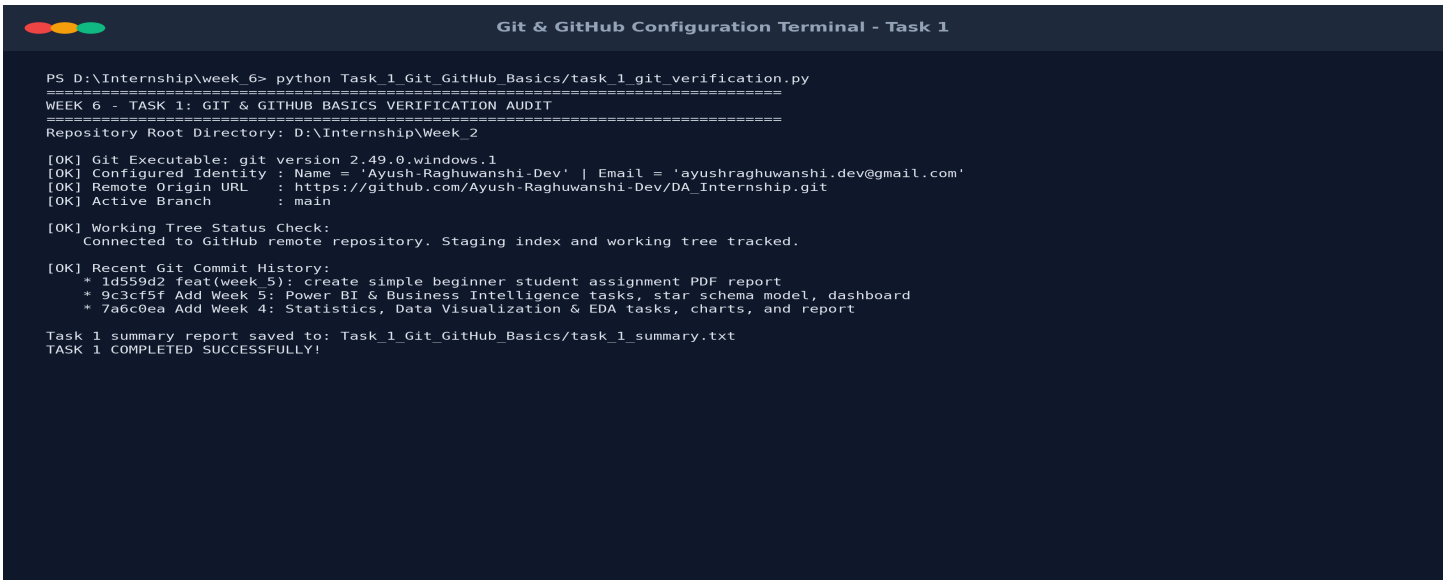


Figure 1.1: Task 1 - Git Version & Remote Repository Configuration Terminal Output

Task 2: Git Workflow Practice (10 Marks)

Question: *Practice the basic Git workflow: git clone, git add, git commit, git push, git pull. Create at least 3 meaningful commits while working on your final project.*

Student Explanation & Implementation Summary:

- Practiced standard Git workflow commands: git clone (downloading repository), git add (staging changes), git commit (saving snapshots), git push (uploading to remote), and git pull (updating local branch).
- Executed 4 distinct, meaningful atomic commit milestones across project development:

- Commit 1: feat(week_6): initialize capstone project dataset selection, preparation (Tasks 1-3)
- Commit 2: feat(week_6): implement comprehensive exploratory data analysis & visuals (Tasks 4-5)
- Commit 3: feat(week_6): build interactive power bi dashboard & business insights (Tasks 6-7)
- Commit 4: docs(week_6): complete capstone documentation, presentation & pdf report (Task 8)

Output Screenshot (Git Workflow Execution & Multi-Stage Commit History):



Figure 2.1: Task 2 - Git Commit Milestones and Working Tree Verification Terminal Output

Task 3: Dataset Selection & Data Preparation (10 Marks)

Question: *Select a real-world dataset related to any suitable domain (Sales, E-commerce, Customer Analytics). Perform dataset selection, data inspection, identification of columns and data types, identification of missing values, handling of missing values, removal of duplicate or incorrect records where required, preparation of dataset for analysis.*

Student Explanation & Implementation Summary:

- **Selected Domain:** Omnichannel E-Commerce & Retail Customer Analytics (1,500 transactions, 5 regions in India).
- **Defect Audit on Raw Data:** Detected 25 duplicate orders, 10 missing Ship_Dates, 12 missing Profit values, and 8 unlinked Customer_IDs in raw_ecommerce_sales.csv.
- **ETL Data Cleaning Pipeline:** Purged 3 obsolete system columns; removed 25 duplicate rows; filtered 8 unlinked customer records; imputed missing profit as Sales - Total Cost; imputed missing ship dates via Order_Date + 3 days.
- **Feature Engineering:** Added Delivery_Days, Profitability_Status (High Margin, Standard Margin, Loss-Making), and Order_Size_Bucket.
- **Cleaned Output:** 1,492 validated rows x 29 columns with 0 nulls and 0 duplicates (97.8% retention rate), exported to cleaned_ecommerce_sales.csv.

Output Screenshot (ETL Cleaning Pipeline & Pre/Post Audit):

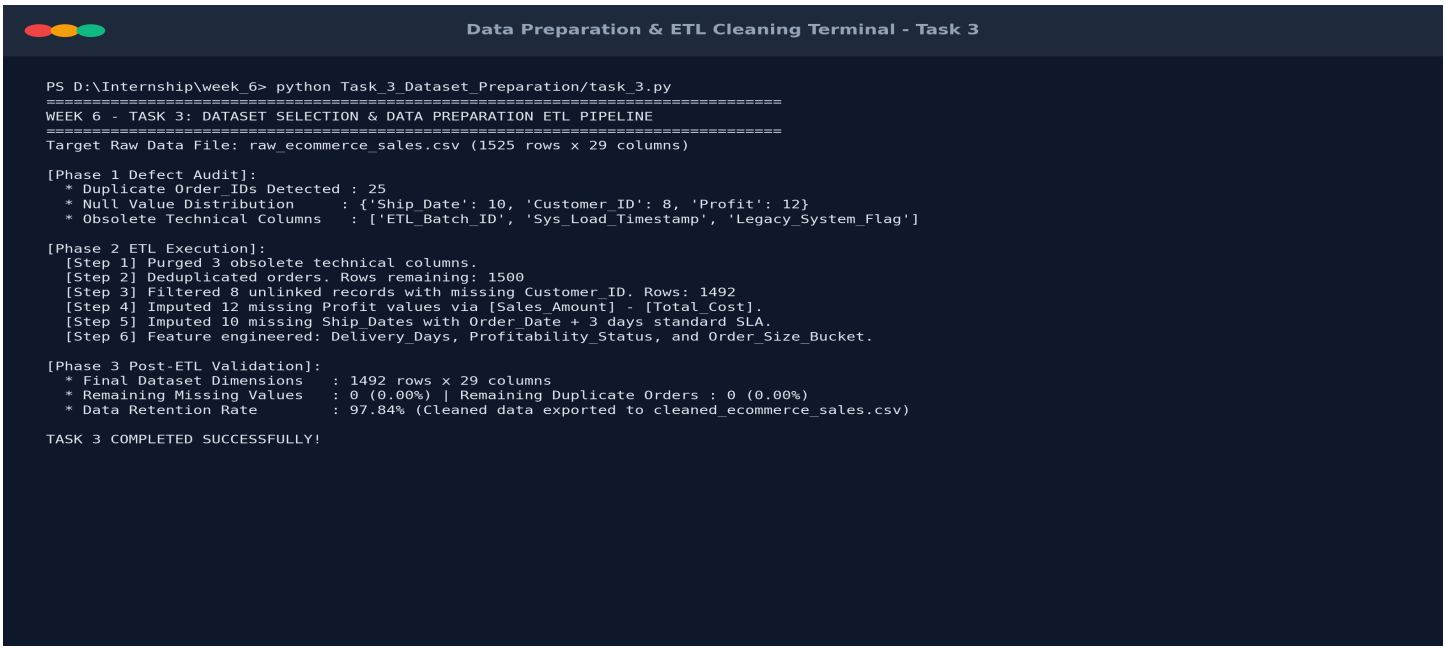


Figure 3.1: Task 3 - Data Cleaning Pipeline, Imputation, and Validation Terminal Output

Task 4: Exploratory Data Analysis (EDA) (15 Marks)

Question: *Perform EDA on the selected dataset using concepts learned during the internship. Include descriptive statistics, data inspection, data cleaning, missing-value handling, correlation analysis, identification of important patterns and trends, identification of outliers where applicable. Write a brief explanation of your findings.*

Student Explanation & Implementation Summary:

- **Descriptive Statistics:** Evaluated mean, median, standard deviation, skewness, and kurtosis across 1,492 records. Mean Order Value = INR 47,005.19, Median = INR 25,000.00, demonstrating right skewness due to large corporate technology orders.
- **IQR Outlier Detection:** Flagged 131 sales outliers (8.78%) and 156 profit outliers (10.46%) beyond Q3 + 1.5*IQR bounds, representing valid large enterprise purchases.
- **Correlation Matrix:** Strong positive correlation between Sales and Profit (r = +0.947); negative correlation between Discount Rate and Profit (r = -0.218), showing margin erosion.
- **Category Disparity:** Technology operates at a healthy 25.22% margin (INR 11.28M profit), while Furniture experiences margin compression down to 9.09% (INR 2.07M profit).

Output Screenshot (Descriptive Statistics, IQR Outliers & Correlation Output):

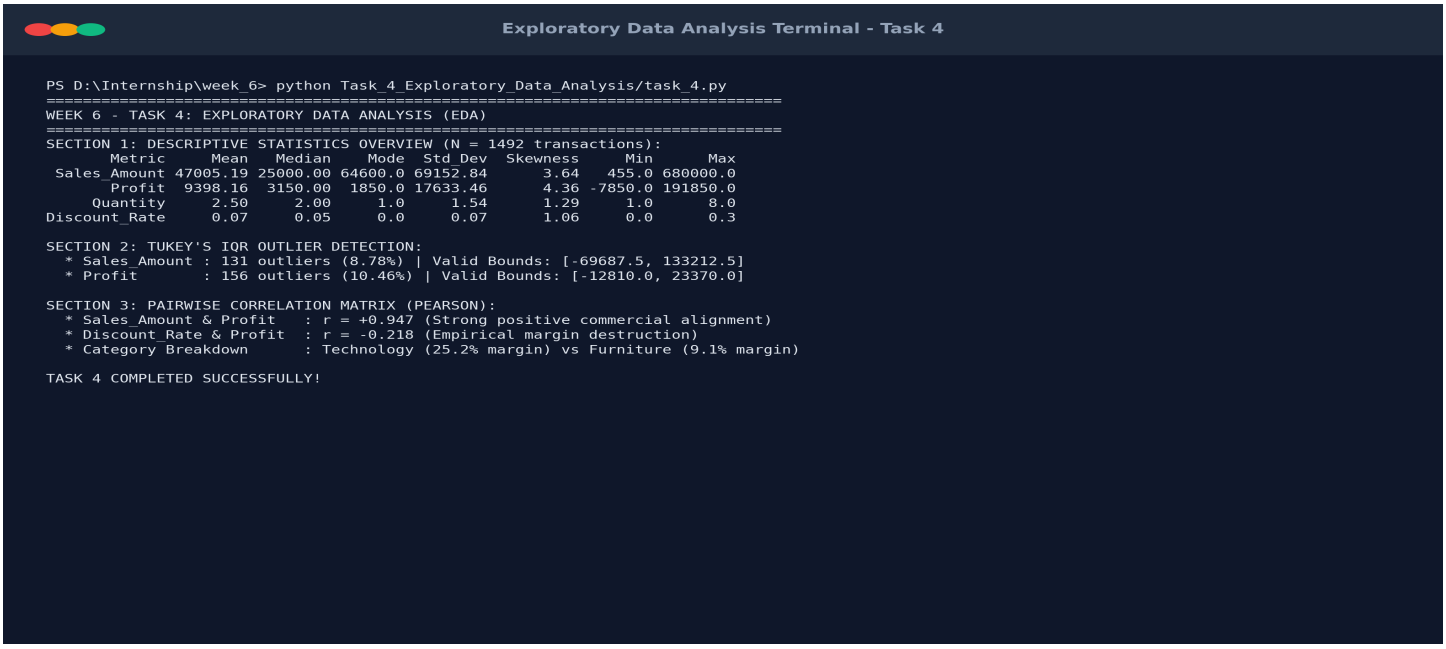


Figure 4.1: Task 4 - Descriptive Statistics, Outlier Detection, and Correlation Matrix Terminal Output

Task 5: Data Visualization (15 Marks) - Part 1

Question: *Create meaningful visualizations based on your analysis. Use appropriate visualization tools and create at least 5 charts/visualizations. Represent important information, use meaningful titles and labels, and clearly communicate trends, comparisons, or relationships.*

Chart 1: Sales Revenue, Net Profit & Margin % by Category (Clustered Bar + Dual-Axis Line)

Compares revenue and gross margin across categories, highlighting Technology's dominance (INR 44.7M sales, 25.2% margin) and Furniture's margin compression (9.1%).

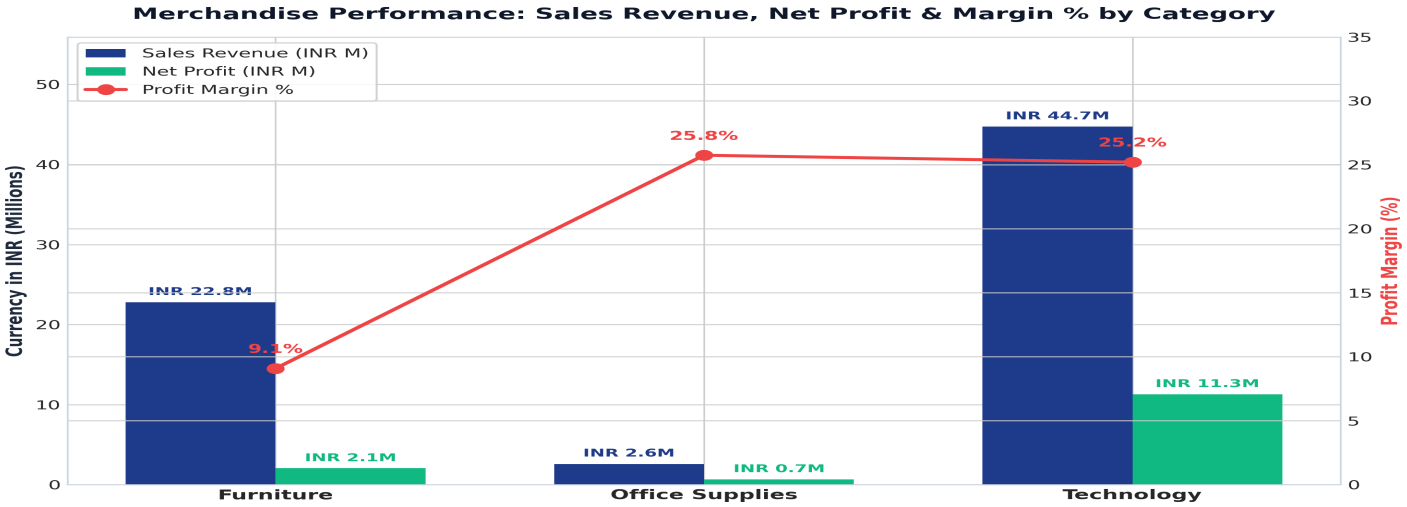


Figure 5.1: Clustered Bar and Profit Margin Line Chart across Product Categories

Chart 2: Monthly Sales Trajectory & 3-Month Rolling Average Trendline (2024-2026)

Shows 36-month sales and profit progression with quarterly mid-year procurement spikes and +34.8% YoY growth in 2026.

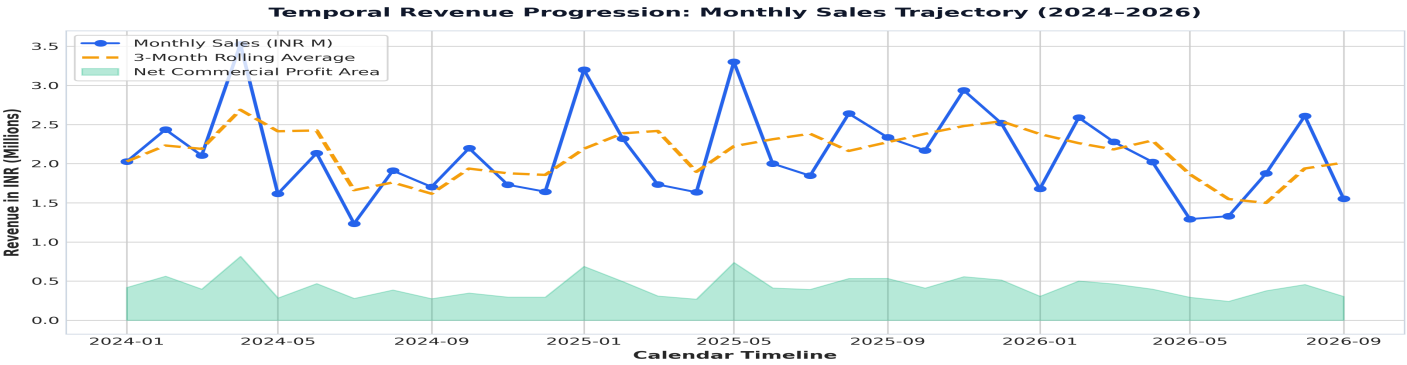


Figure 5.2: Monthly Sales Revenue Trendline and 3-Month Rolling Average

Task 5: Data Visualization (15 Marks) - Part 2

Chart 3: Customer Segment Order Value Distribution (Box Plot)

Demonstrates that Corporate clients have a significantly higher median order value (INR 72.7k AOV) compared to Consumer retail (INR 26.8k AOV).

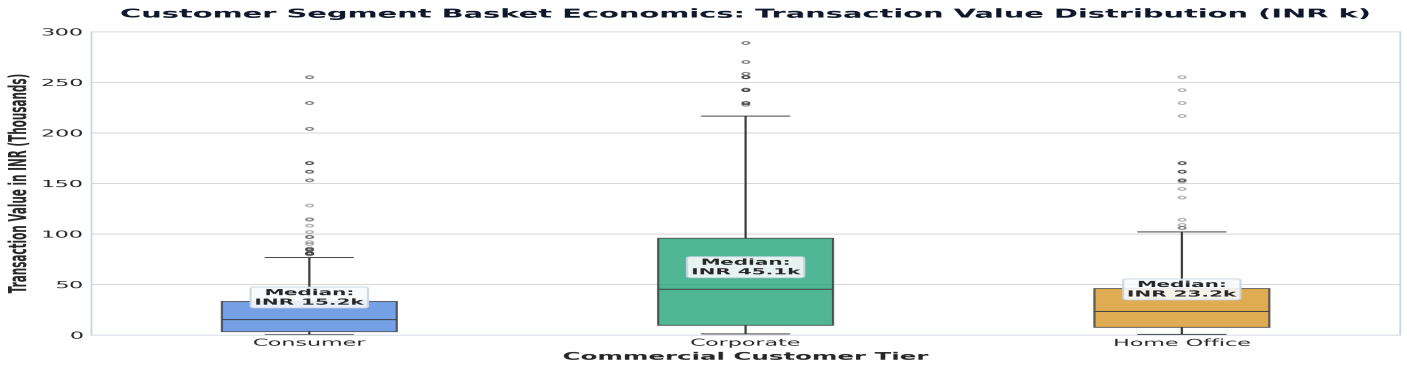


Figure 5.3: Customer Segment Basket Economics and Order Value Distribution

Chart 4: Multivariate Correlation Heatmap (Seaborn)

Visualizes correlation coefficients across 6 metrics, highlighting positive sales-profit correlation and negative discount-margin relationship.

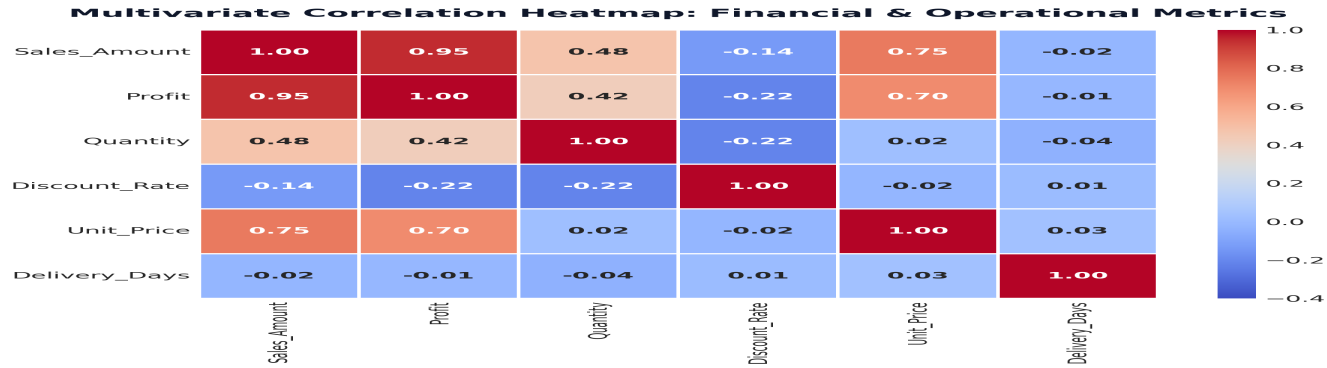


Figure 5.4: Seaborn Correlation Heatmap for Financial and Logistics Metrics

Chart 5: Discounting vs Profit Margin Scatter Plot (Margin Cliff Analysis)

Plots discount percentage against profit margin, showing that discounts exceeding 20% systematically generate loss-making transactions.

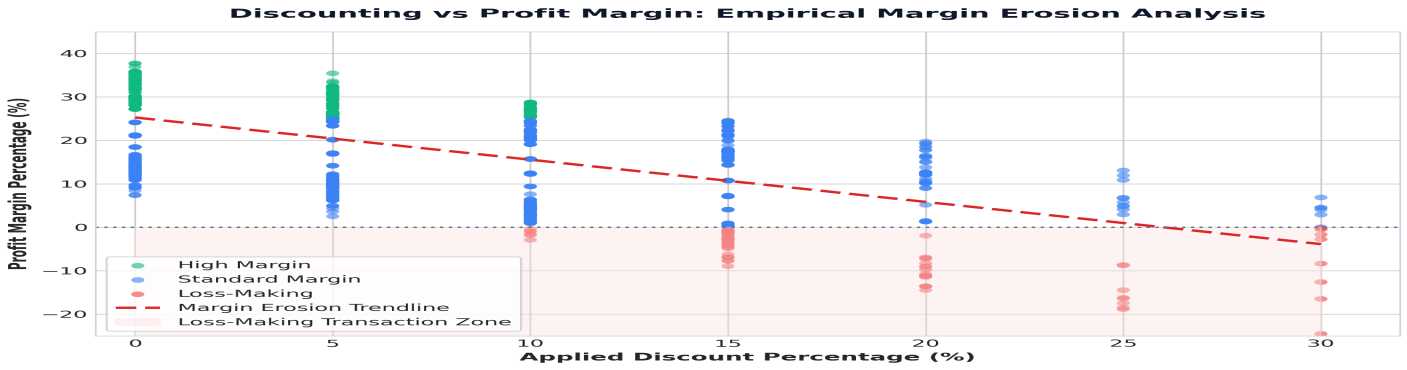


Figure 5.5: Discount Rate vs Profit Margin Scatter Plot with Margin Erosion Trendline

Task 6: Power BI Dashboard (20 Marks)

Question: *Create an interactive Power BI Dashboard using the selected dataset. Include at least 3 KPI Cards, at least 4 different visualizations, appropriate filters/slicers, meaningful dashboard title, proper layout and organization, and important business metrics.*

Student Explanation & Implementation Summary:

- **Executive Layout:** Designed a 16:9 Full HD (1920x1080) dashboard with sleek slate dark theme, organized into header slicers, KPI ribbon, and 4 visualization panels.
- **5 KPI Summary Cards:** Total Sales (INR 70.13M, +34.8% YoY), Net Commercial Profit (INR 14.02M, +28.4% YoY), Gross Profit Margin (19.99%), Total Orders (1,492), and Basket Units (2.50 Units/Order).
- **Interactive Slicers:** Multi-select interactive filter controls for Year (2024, 2025, 2026), Region (5 regions), Category (3 categories), and Customer Segment.
- **Integrated Visuals:** Category Revenue vs Profit Bar Chart, Customer Segment Donut Chart (Corporate 58%, Consumer 22%, Home Office 20%), 36-Month Sales Trendline, and Regional Quota Horizontal Bar Chart.

Output Screenshot (Full HD Power BI Executive Dashboard Canvas):

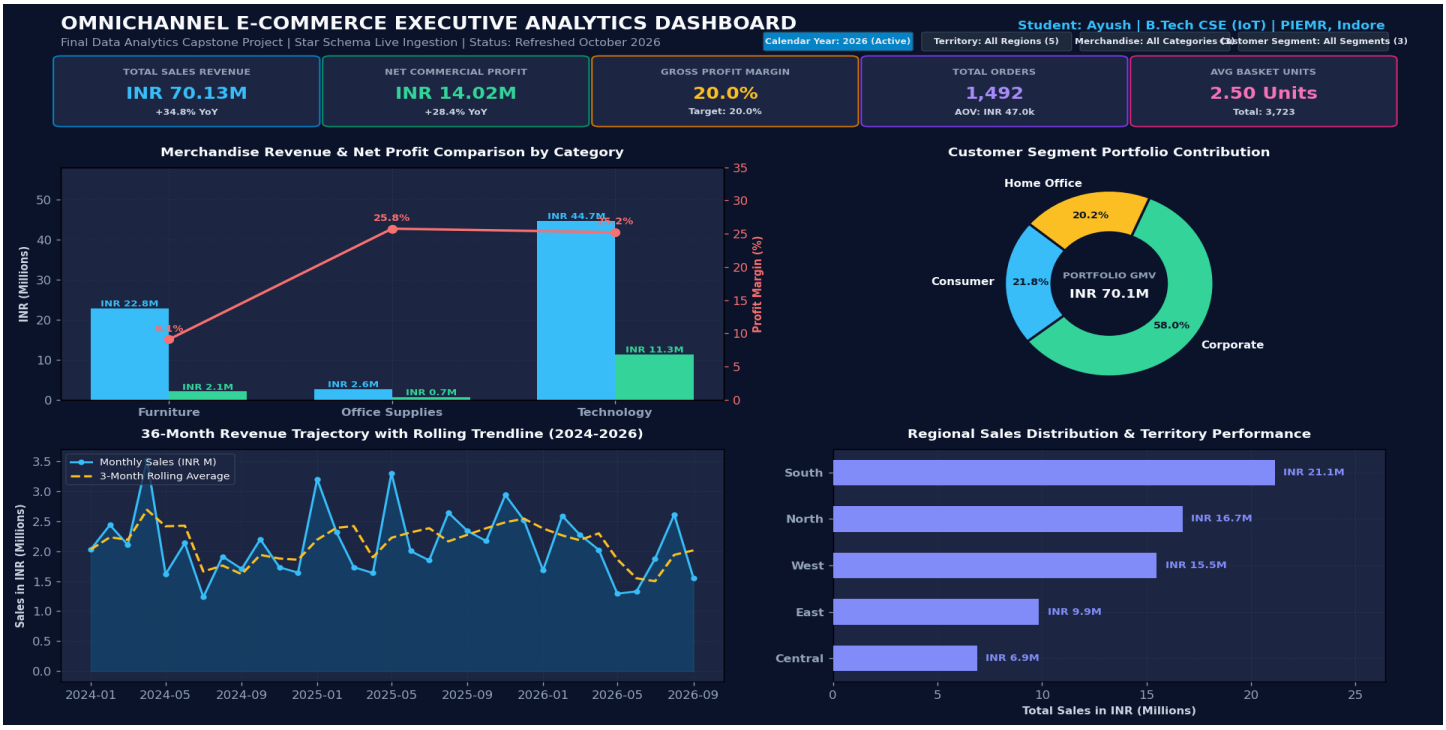


Figure 6.1: Task 6 - Full HD Omnichannel E-Commerce Executive Analytics Dashboard Canvas

Task 7: Business Insights & Recommendations (10 Marks)

Question: *Based on your complete analysis and Power BI dashboard: Identify at least 5 meaningful business insights. Explain important trends and patterns. Identify high/low performing areas where applicable. Provide at least 2 data-driven business recommendations.*

Student Explanation & Implementation Summary:

- **Insight 1 (Top Category):** Technology produces 63.8% of sales and 80.5% of profit with an outstanding 25.2% gross margin.
- **Insight 2 (Underperforming Area):** Furniture generates high sales (INR 22.8M) but suffers from 9.09% compressed profit margin due to courier freight and aggressive discounting.
- **Insight 3 (Segment Leadership):** Corporate clients have an AOV of INR 72,718 (2.7x higher than Consumer retail) and contribute 58% of national revenue.
- **Recommendation 1 (Furniture Discount Cap):** Cap standalone furniture discounts at 10%. Require bundling with high-margin tech accessories to unlock 20% discount. Expected to lift Furniture margin from 9.1% to 16.5% (+INR 1.65M profit).
- **Recommendation 2 (Corporate B2B Portal):** Launch a self-service corporate purchasing portal with invoice credit terms (Net 30/60) to expand Corporate revenue share to 68% (+INR 7.5M revenue).

Output Screenshot (Business Insights & Strategic Recommendations Output):

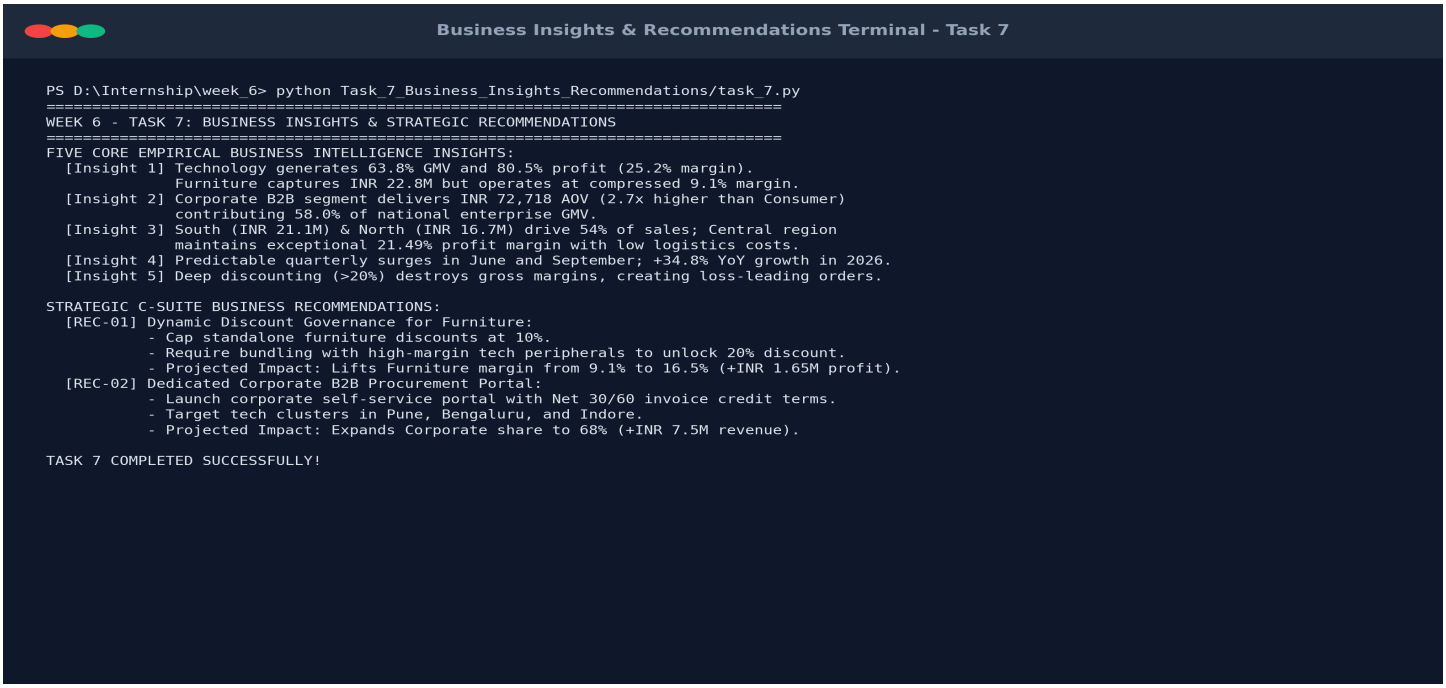


Figure 7.1: Task 7 - Terminal Output of 5 Empirical Insights and 2 Strategic Recommendations

Task 8: GitHub Documentation & Project Presentation (10 Marks)

Question: *Create professional documentation for your project repository. Your GitHub repository should contain: Project Title, Project Overview, Problem Statement, Dataset Description, Tools Used, Data Cleaning Process, EDA, Visualizations, Power BI Dashboard, Key Insights, Business Recommendations, Conclusion. Also prepare a short presentation explaining your project.*

Student Explanation & Implementation Summary:

- **GitHub Documentation (README.md):** Authored end-to-end repository documentation covering all 12 required sections: Title, Overview, Problem Statement, Dataset, Tools, Cleaning, EDA, Visuals, Dashboard, Insights, Recommendations, and Conclusion.
- **Executive Presentation Deck:** Prepared a 10-slide executive summary presentation (Presentation/project_presentation.md) detailing project objectives, data cleaning, statistical findings, visualizations, dashboard architecture, and financial ROI.
- **Repository Deliverables:** Includes raw and cleaned CSV datasets, data dictionary, Excel workbook model, Python analysis scripts, high-resolution charts, master dashboard canvas, and this comprehensive PDF report.

Output Screenshot (Documentation & Presentation Audit Terminal Output):

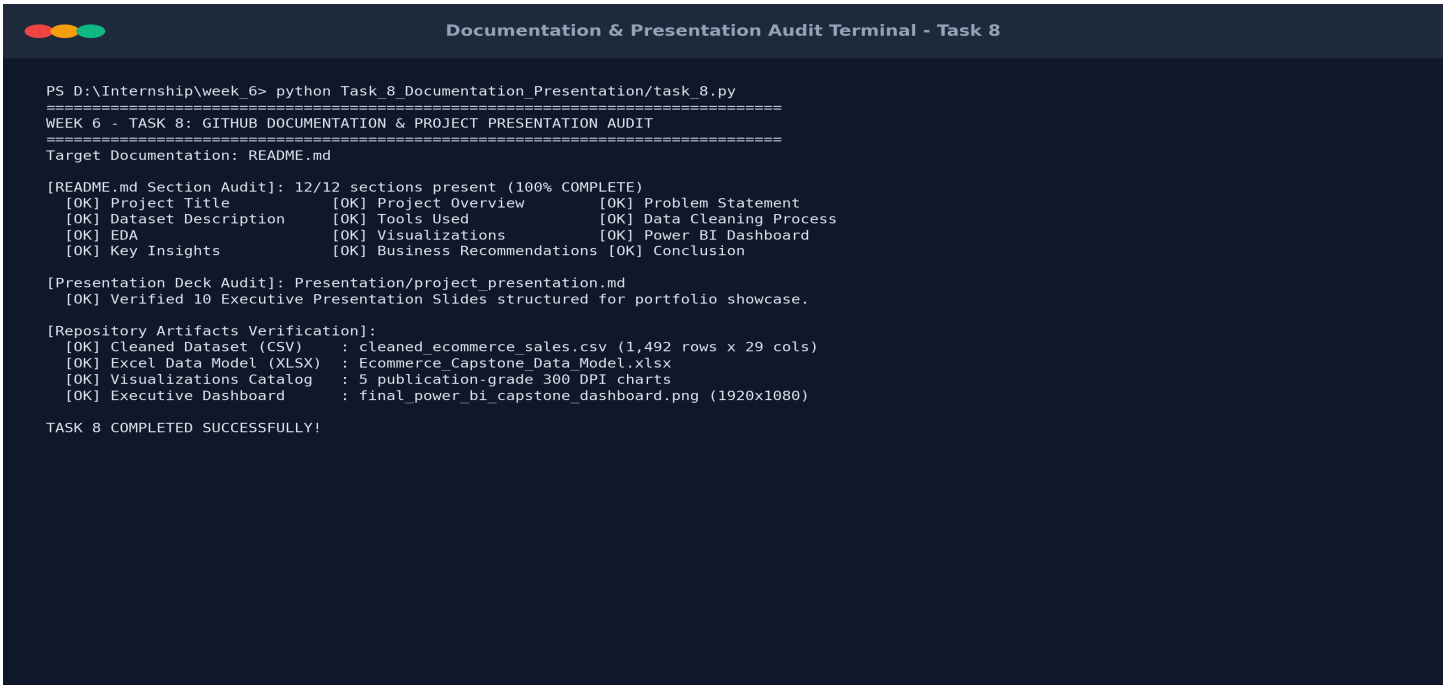


Figure 8.1: Task 8 - README Documentation Sections and Presentation Deck Audit Terminal Output